The inverse function theorem for everywhere differentiable maps

Terence Tao · 12 September 2011

https://terrytao.wordpress.com/2011/09/12/the-inverse-function-theorem-for-everywhere-differentiable-maps/

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The classical inverse function theorem reads as follows:

> **Theorem 1 (C^1 inverse function theorem)** Let $\Omega \subset \mathbf{R}^n$ be an open set, and let $f : \Omega \to \mathbf{R}^n$ be an continuously differentiable function, such that for every $x_0 \in \Omega$, the derivative map $Df(x_0) : \mathbf{R}^n \to \mathbf{R}^n$ is invertible. Then f is a local homeomorphism; thus, for every $x_0 \in \Omega$, there exists an open neighbourhood U of x_0 and an open neighbourhood V of $f(x_0)$ such that f is a homeomorphism from U to V.

It is also not difficult to show by inverting the Taylor expansion

$$f(x) = f(x_0) + Df(x_0)(x - x_0) + o(\|x - x_0\|)$$

that at each x_0, the local inverses $f^{-1} : V \to U$ are also differentiable at $f(x_0)$ with derivative

$$Df^{-1}(f(x_0)) = Df(x_0)^{-1}. \tag{1}$$

The textbook proof of the inverse function theorem proceeds by an application of the contraction mapping theorem. Indeed, one may normalise $x_0 = f(x_0) = 0$ and $Df(0)$ to be the identity map; continuity of Df then shows that $Df(x)$ is close to the identity for small x, which may be used (in conjunction with the fundamental theorem of calculus) to make $x \mapsto x - f(x) + y$ a contraction on a small ball around the origin for small y, at which point the contraction mapping theorem readily finishes off the problem.

I recently learned (after I asked this question on Math Overflow) that the hypothesis of continuous differentiability may be relaxed to just everywhere differentiability:

> **Theorem 2 (Everywhere differentiable inverse function theorem)** Let $\Omega \subset \mathbf{R}^n$ be an open set, and let $f : \Omega \to \mathbf{R}^n$ be an everywhere differentiable function, such that for every $x_0 \in \Omega$, the derivative map $Df(x_0) : \mathbf{R}^n \to \mathbf{R}^n$ is invertible. Then f is a local homeomorphism; thus, for every $x_0 \in \Omega$, there exists an open neighbourhood U of x_0 and an open neighbourhood V of $f(x_0)$ such that f is a homeomorphism from U to V.

As before, one can recover the differentiability of the local inverses, with the derivative of the inverse given by the usual formula (1).

This result implicitly follows from the more general results of Cernavskii about the structure of finite-to-one open and closed maps, however the arguments there are somewhat complicated (and subsequent proofs of those results, such as the one by Vaisala, use some powerful tools from algebraic geometry, such as dimension theory). There is however a more elementary proof of Saint Raymond that was pointed out to me by Julien Melleray. It only uses basic point-set topology (for instance, the concept of a connected component) and the basic topological and geometric structure of Euclidean space (in particular relying primarily on local compactness, local connectedness, and local convexity). I decided to present (an arrangement of) Saint Raymond's proof here.

To obtain a local homeomorphism near x_0, there are basically two things to show: local surjectivity near x_0 (thus, for y near $f(x_0)$, one can solve $f(x) = y$ for some x near x_0) and local injectivity near x_0 (thus, for distinct x_1, x_2 near $f(x_0)$, $f(x_1)$ is not equal to $f(x_2)$). Local surjectivity is relatively easy; basically, the standard proof of the inverse function theorem works here, after replacing the contraction

mapping theorem (which is no longer available due to the possibly discontinuous nature of Df) with the Brouwer fixed point theorem instead (or one could also use degree theory, which is more or less an equivalent approach). The difficulty is local injectivity – one needs to preclude the existence of nearby points x_1, x_2 with $f(x_1) = f(x_2) = y$; note that in contrast to the contraction mapping theorem that provides both existence and uniqueness of fixed points, the Brouwer fixed point theorem only gives existence and not uniqueness.

In one dimension $n = 1$ one can proceed by using Rolle's theorem. Indeed, as one traverses the interval from x_1 to x_2, one must encounter some intermediate point x_* which maximises the quantity $|f(x_*) - y|$, and which is thus instantaneously non-increasing both to the left and to the right of x_*. But, by hypothesis, $f'(x_*)$ is non-zero, and this easily leads to a contradiction.

Saint Raymond's argument for the higher dimensional case proceeds in a broadly similar way. Starting with two nearby points x_1, x_2 with $f(x_1) = f(x_2) = y$, one finds a point x_* which "locally extremises" $\|f(x_*) - y\|$ in the following sense: $\|f(x_*) - y\|$ is equal to some $r_* > 0$, but x_* is adherent to at least two distinct connected components U_1, U_2 of the set $U = \{x : \|f(x) - y\| < r_*\}$. (This is an oversimplification, as one has to restrict the available points x in U to a suitably small compact set, but let us ignore this technicality for now.) Note from the non-degenerate nature of $Df(x_*)$ that x_* was already adherent to U; the point is that x_* "disconnects" U in some sense. Very roughly speaking, the way such a critical point x_* is found is to look at the sets $\{x : \|f(x) - y\| \leq r\}$ as r shrinks from a large initial value down to zero, and one finds the first value of r_* below which this set disconnects x_1 from x_2. (Morally, one is performing some sort of Morse theory here on the function $x \mapsto \|f(x) - y\|$, though this function does not have anywhere near enough regularity for classical Morse theory to apply.)

The point x_* is mapped to a point $f(x_*)$ on the boundary $\partial B(y, r_*)$ of the ball $B(y, r_*)$, while the components U_1, U_2 are mapped to the interior of this ball. By using a continuity argument, one can show (again very roughly speaking) that $f(U_1)$ must contain a "hemispherical" neighbourhood $\{z \in B(y, r_*) : \|z - f(x_*)\| < \kappa\}$ of $f(x_*)$ inside $B(y, r_*)$, and similarly for $f(U_2)$. But then from differentiability of f at x_*, one can then show that U_1 and U_2 overlap near x_*, giving a contradiction.

The rigorous details of the proof are provided below the fold.

— 1. Proof —

Fix $x_0 \in \Omega$. By a translation, we may assume $x_0 = f(x_0) = 0$; by a further linear change of variables, we may also assume $Df(0)$ (which by hypothesis is non-singular) to be the identity map. By differentiability, we have

$$f(x) = x + o(\|x\|)$$

as $x \to 0$. In particular, there exists a ball $B(0, r_0)$ in Ω such that

$$\|f(x) - x\| < \frac{1}{2}\|x\|$$

for all $x \in B(0, r_0)$; by rescaling we may take $r_0 = 1$, thus

$$\|f(x) - x\| < \frac{1}{2} \| x \| \text{ whenever } \|x\| \leq 1. \tag{2}$$

Among other things, this gives a uniform lower bound

$$\|f(x)\| > \frac{1}{2} \tag{3}$$

for all $x \in \partial B(0, 1)$, and a uniform upper bound

$$\|f(x)\| < \frac{1}{10} \tag{4}$$

for all $x \in B(0, \frac{1}{20})$; thus f maps $B(0, \frac{1}{20})$ to $B(0, \frac{1}{10})$.

Proposition 3 (Local surjectivity) For any $0 < r < 1$, $f(B(0,r))$ contains $B(0,r/2)$.

Proof: Let $y \in B(0,r/2)$. From (2), we see that the map $f : \partial B(0,r) \to f(\partial B(0,r))$ avoids y, and has degree 1 around y; contracting $\partial B(0,r)$ to a point, we conclude that $f(x) = y$ for some $x \in B(0,r)$, yielding the claim.

Alternatively, one may proceed by invoking the Brouwer fixed point theorem, noting that the map $x \mapsto x - f(x) + y$ is continuous and maps the closed ball $\overline{B(0,r)}$ to the open ball $B(0,r)$ by (2), and has a fixed point precisely when $f(x) = y$.

A third argument (avoiding the use of degree theory or the Brouwer fixed point theorem, but requiring one to replace $B(0,r/2)$ with the slightly smaller ball $B(0,r/3)$) is as follows: let $x \in \overline{B(0,r)}$ minimise $\|f(x) - y\|$. From (2) and the hypothesis $y \in B(0,r/3)$ we see that x lies in the interior $B(0,r)$. If the minimum is zero, then we have found a solution to $f(x) = y$ as required; if not, then we have a stationary point of $x \mapsto \|f(x) - y\|$, which implies that $Df(x)$ is degenerate, a contradiction. (One can recover the full ball $B(0,r/2)$ by tweaking the expression $\|f(x) - y\|$ to be minimised in a suitable fashion; we leave this as an exercise for the interested reader.) □

Corollary 4 f is an open map: the image of any open set is open.

Proof: It suffices to show that for every $x \in \Omega$, the image of any open neighbourhood of x is an open neighbourhood of $f(x)$. Proposition 3 handles the case $x = 0$; the general case follows by renormalising. □

Suppose we could show that f is injective on $B(0, \frac{1}{20})$. By Corollary 4, the inverse map $f^{-1} : f(B(0, \frac{1}{20})) \to B(0, \frac{1}{20})$ is also continuous. Thus f is a homeomorphism from $B(0, \frac{1}{20})$ to $f(B(0, \frac{1}{20}))$, which are both neighbourhoods of 0 by Proposition 3; giving the claim.

It remains to establish injectivity. Suppose for sake of contradiction that this was not the case. Then there exists $x_1, x_2 \in B(0, \frac{1}{20})$ and $y \in B(0, \frac{1}{10})$ such that

$$y = f(x_1) = f(x_2).$$

For every radius $r \geq 0$, the set

$$K_r := \{x \in \Omega : \|f(x) - y\| \leq r\}$$

is closed and contains both x_1 and x_2. Let K_r^1 denote the connected component of K_r that contains x_1. Since K_r is non-decreasing in r, K_r^1 is non-decreasing also.

Now let us study the behaviour of K_r^1 as r ranges from 0 to $\frac{4}{10}$. The two extreme cases are easy to analyse:

Lemma 5 $K_0^1 = \{x_1\}$.

Proof: Since $Df(x_1)$ is non-singular, we see from differentiability that $f(x) \neq f(x_1)$ for all $x \neq x_1$ sufficiently close to x_1. Thus x_1 is an isolated point of K_0, and the claim follows. □

Lemma 6 We have $B(0, \frac{1}{20}) \subset K_r^1 \subset B(0,1)$ for all $\frac{2}{10} \leq r \leq \frac{4}{10}$. In particular, K_r^1 is compact for all $0 \leq r \leq \frac{4}{10}$, and contains x_2 for $\frac{2}{10} \leq r \leq \frac{4}{10}$.

Proof: Since $f(B(0, \frac{1}{20})) \subset B(f(0), \frac{1}{10}) \subset \overline{B(y,r)}$, we see that $B(0, \frac{1}{20}) \subset K_r$; since $B(0, \frac{1}{20})$ is connected and contains x_1, we conclude that $B(0, \frac{1}{20}) \subset K_r^1$.

Next, if $x \in \partial B(0,1)$, then by (3) we have $f(x) \notin B(0, \frac{1}{2})$, and hence $f(x) \notin \overline{B(y,r)}$. Thus K_r is disjoint from the sphere $\partial B(0,1)$. Since x_1 lies in the interior of this sphere we thus have $K_r^1 \subset B(0,1)$ as required. □

Next, we show that the K_r^1 increase continuously in r:

Lemma 7 If $0 \leq r < \frac{1}{20}$ and $\epsilon > 0$, then for $r < r' < \frac{1}{20}$ sufficiently close to r, $K_{r'}^1$ is contained in an ϵ-neighbourhood of K_r^1.

Proof: By the finite intersection property, it suffices to show that $\bigcap_{r'>r} K^1_{r'} = K^1_r$. Suppose for contradiction that there is a point x outside of K^1_r that lies in $K^1_{r'}$ for all $r' > r$. Then x lies in $K_{r'}$ for all $r' > r$, and hence lies in $K_r \cap B(0,1)$. As x and x_1 lie in different connected components of the compact set $K_r \cap \overline{B(0,1)}$ (recall that K_r is disjoint from $\partial B(0,1)$), there must be a partition of $K_r \cap \overline{B(0,1)}$ into two disjoint closed sets F, G that separate x from x_1 (for otherwise the only clopen sets in $K_r \cap \overline{B(0,1)}$ that contain x_1 would also contain x, and their intersection would then be a connected subset of $K_r \cap \overline{B(0,1)}$ that contains both x_1 and x, contradicting the fact that x lies outside K^1_r). By normality, we may find open neighbourhoods U, V of F, G that are disjoint. For all x on the boundary ∂U, one has $\|f(x) - y\| > r$ for all $x \in \partial U$. As ∂U is compact and f is continuous, we thus have $\|f(x) - y\| > r'$ for all $x \in \partial U$ if r' is sufficiently close to r. This makes $U \cap K_{r'}$ clopen in $K_{r'}$, and so x cannot lie in $K^1_{r'}$, giving the desired contradiction. □

Observe that K^1_r contains x_2 for $r \geq \frac{2}{10}$, but does not contain x_2 for $r = 0$. By the monotonicity of the K^1_r and least upper bound principle, there must therefore exist a critical $0 \leq r_* \leq \frac{2}{10}$ such that K^1_r contains x_2 for all $r > r_*$, but does not contain x_2 for $r < r_*$. From Lemma 7 we see that $K^1_{r_*}$ must also contain x_2. In particular, by Lemma 5, $r_* > 0$.

We now analyse the critical set $K^1_{r_*}$. By construction, this set is connected, compact, contains both x_1 and x_2, contained in $B(0,1)$, and one has $\|f(x) - y\| \leq r_*$ for all $x \in K^1_{r_*}$.

Lemma 8 The set $U := \{x \in K^1_{r_*} : \|f(x) - y\| < r_*\}$ is open and disconnected.

Proof: The openness is clear from the continuity of f (and the local connectedness of $\mathbf{R}^n$). Now we show disconnectedness. Being an open subset of $\mathbf{R}^n$, connectedness is equivalent to path connectedness, and x_1 and x_2 both lie in U, so it suffices to show that x_1 and x_2 cannot be joined by a path γ in U. But if such a path γ existed, then by compactness of γ and continuity of f, one would have $\gamma \subset K_r$ for some $r < r_*$. This would imply that $x_2 \in K^1_r$, contradicting the minimal nature of r_*, and the claim follows.

Lemma 9 U has at most finitely many connected components.

Proof: Let U_1 be a connected component of U; then $f(U_1)$ is non-empty and contained in $B(y, r_*)$. As U is open, U_1 is also open, and thus by Corollary 4, $f(U_1)$ is open also.

We claim that $f(U_1)$ is in fact all of $B(y, r_*)$. Suppose this were not the case. As $B(y, r_*)$ is connected, this would imply that $f(U_1)$ is not closed in $B(y, r_*)$; thus there is an element z of $B(y, r_*)$ which is adherent to $f(U_1)$, but does not lie in $f(U_1)$. Thus one may find a sequence x_n in U_1 with $f(x_n)$ converging to z. By compactness of $K^1_{r_*}$ (which contains U_1), we may pass to a subsequence and assume that x_n converges to a limit x in $K^1_{r_*}$; then $f(x) = z$. By continuity, there is thus a ball B centred at x that is mapped to $B(y, r)$ for some $r < r_*$; this implies that B lies in K_{r_*} and hence in $K^1_{r_*}$ (since $x \in K^1_{r_*}$) and thence in U (since r is strictly less than r_*). As x is adherent to U_1 and B is connected, we conclude that B lies in U_1. In particular x lies in U_1 and so $z = f(x)$ lies in $f(U_1)$, a contradiction.

As $f(U_1)$ is equal to $B(y, r_*)$, we thus see that U_1 contains an element of $f^{-1}(\{y\})$. However, each element x of $f^{-1}(\{y\})$ must be isolated since $Df(x)$ is non-singular. By compactness of $K^1_{r_*}$, the set $K^1_{r_*}$ (and hence U) thus contains at most finitely many elements of $f^{-1}(\{y\})$, and so there are finitely many components as claimed. □

Lemma 10 Every point in $K^1_{r_*}$ is adherent to U (i.e. $\overline{U} = K^1_{r_*}$).

Proof: If $x \in K^1_{r_*}$, then $\|f(x) - y\| \leq r_*$. If $\|f(x) - y\| < r_*$ then $x \in U$ and we are done, so we may assume $\|f(x) - y\| = r_*$. By differentiability, one has

$$f(x') = f(x) + Df(x)(x' - x) + o(\|x' - x\|)$$

for all x' sufficiently close to x. If we choose x' to lie on a ray emenating from x such that $Df(x)(x' - x)$ lies on a ray pointing towards y from $f(x)$ (this is possible as $Df(x)$ is non-singular), we conclude that for all x' sufficiently close to x on this ray, $\|f(x') - y\| < r_*$. Thus all such points x' lie in K_{r_*};

since x lies in $K^1_{r_*}$ and the ray is locally connected, we see that all such points x' in fact lie in $K^1_{r_*}$ and thence in U. The claim follows. □

> **Corollary 11** There exists a point $x_* \in K^1_{r_*}$ with $\|f(x_*) - y\| = r_*$ (i.e. x_* lies outside U) which is adherent to at least two connected components of U.

Proof: Suppose this were not the case, then the closures of all the connected components of U would be disjoint. (Note that an element of one connected component of U cannot lie in the closure of another component.) By Lemma 10, these closures would form a partition of $K^1_{r_*}$ by closed sets. By Lemma 8, there are at least two such closed sets, each of which is non-empty; by Lemma 9, the number of such closed sets is finite. But this contradicts the connectedness of $K^1_{r_*}$. □

Next, we prove

> **Proposition 12** Let $x_* \in K^1_{r_*}$ be such that $\|f(x_*) - y\| = r_*$, and suppose that x is adherent to a connected component U_1 of U. Let ω be the vector such that
>
> $$Df(x_*)\omega = y - f(x_*) \tag{5}$$
>
> (this vector exists and is non-zero since $Df(x_*)$ is non-singular). Then U_1 contains an open ray of the form $\{x_* + t\omega : 0 < t < \epsilon\}$ for some $\epsilon > 0$.

This together with Corollary 11 gives the desired contradiction, since one cannot have two distinct components U_1, U_2 both contain a ray from x_* in the direction ω.

Proof: As f is differentiable at x_*, we have

$$f(x_* + t\omega) = f(x_*) + Df(x_*)t\omega + o(|t|)$$

for all sufficiently small t; we rearrange this using (5) as

$$f(x_* + t\omega) - y = (1-t)(f(x_*) - y) + o(|t|).$$

In particular, $f(x_* + t\omega) \in B(y, r_*)$ for all sufficiently small positive t. This shows that all sufficiently small open rays $\{x_* + t\omega : 0 < t < \epsilon\}$ lie in K_{r_*}, hence in $K^1_{r_*}$ (since $x_* \in K^1_{r_*}$), and hence in U. In fact, the same argument shows that there is a cone

$$\{x_* + t\omega' : 0 < t < \epsilon; \|\omega' - \omega\| \leq \epsilon\} \tag{6}$$

that will lie in U if ϵ is small enough. As this cone is connected, it thus suffices to show that U_1 intersects this cone.

Let $\delta > 0$ be a small radius to be chosen later. As $Df(x_*)$ is non-singular, we see if δ is small enough that $f(x) \neq f(x_*)$ whenever $\|x - x_*\| = \delta$. By continuity, we may thus find $\kappa > 0$ such that $\|f(x) - f(x_*)\| > \kappa$ whenever $\|x - x_*\| = \delta$.

Consider the set

$$U' := \{x \in U_1 : \|x - x_*\| \leq \delta; \|f(x) - f(x_*)\| < \kappa\}.$$

As x_* is adherent to U_1, U' is non-empty. By construction of κ, we see that we also have

$$U' := \{x \in U_1 : \|x - x_*\| < \delta; \|f(x) - f(x_*)\| < \kappa\}$$

and so U' is open. By Corollary 4, $f(U')$ is then also non-empty and open. By construction, $f(U')$ also lies in the set

$$D := \{z \in B(y, r_*) : \|z - f(x_*)\| < \kappa\}.$$

We claim that $f(U')$ is in fact all of D. The proof will be a variant of the proof of Lemma 9. Suppose this were not the case. As D is connected, this implies that there is an element z of D which is adherent to $f(U')$, but does not lie in $f(U')$. Thus one may find a sequence x_n in U' with $f(x_n)$ converging to z. By compactness of $K^1_{r_*}$ (which contains U'), we may pass to a subsequence and assume that x_n converges to a limit x in $K^1_{r_*}$; then $f(x) = z$. By continuity, there is thus a ball B centred at x contained

in $B(x_*, \delta)$ that is mapped to $B(y, r) \cap D$ for some $r < r_*$; this implies that B lies in K_{r_*} and hence in $K^1_{r_*}$ (since $x \in K^1_{r_*}$) and thence in U (since r is strictly less than r_*). As x is adherent to U_1 and B is connected, we conclude that B lies in U_1 and thence in U'. In particular x lies in U' and so $z = f(x)$ lies in $f(U')$, a contradiction.

As $f(U') = D$, we may thus find a sequence $t_n > 0$ converging to zero, and a sequence $x_n \in U'$, such that

$$f(x_n) = f(x_*) + t_n(y - f(x_*)).$$

However, if δ is small enough, we have $\|f(x_n) - f(x_*)\|$ comparable to $\|x_n - x_*\|$ (cf. (2)), and so x_n converges to x_*. By Taylor expansion, we then have

$$f(x_n) = f(x_*) + Df(x_*)(x_n - x_*) + o(\|x_n - x_*\|)$$

and thus

$$(Df(x_*) + o(1))(x_n - x_*) = t_n Df(x_*)\omega$$

for some matrix-valued error $o(1)$. Since $Df(x_*)$ is invertible, this implies that

$$x_n - x_* = t_n(1 + o(1))\omega = t_n\omega + o(t_n).$$

In particular, x_n lies in the cone (6) for n large enough, and the claim follows. □

Comments (40)



Leandro Cioletti · 12 September 2011

Thanks for sharing this. I read the mathoverflow question and I liked very much this result, but we don't have in our local library a copy of the papers cited there.



Philippe · 12 September 2011

Small correction: the person that pointed this proof to you is Julien Melleray (not Malleray).
[Corrected, thanks – T.]



vineel567 · 12 September 2011

what tools do you use to write such a wonderful technical articles with all the special characters and mathematical style preserved.??? Thanks in advance



Willie Wong · 15 September 2011

Take a look at this page: https://terrytao.wordpress.com/about/ especially after the "Some technical remarks" fold.



Anonymous · 14 September 2011

Sorry I mean everywhere invertibility. Is it necessary? If so why is the local invertibility not sufficient?



Willie Wong · 15 September 2011

The statement does only require local invertibility, no? It only requires invertibility at every point in the open neighborhood Ω. Or are you asking whether invertibility at a single point is sufficient?



Anonymous · 15 September 2011

This result seems counter-intuitive. In the 1D case, if f'(x)<0 for x0 for x>x_0 then it doesn't seem possible for the function f to be invertible at x_0. Also, the derivative is invertible in a neighborhood of x_0 so it seems to satisfy the conditions of the theorem.

[You will need to use < and > instead of < and > to avoid your comments being mangled by the HTML parser. In any event, in one dimension the result is precisely the contrapositive of Rolle's theorem. – T.]



Anonymous · 15 September 2011

It should be f'(x)0 for x > x0.



Josh Swanson · 17 September 2011

Thank you for typing this up! Reading through it was a nice review of basic real analysis and topology.

The Brouwer fixed point theorem is false with open balls, so the map you mention in the proof of Proposition 3 should be between closed balls. Even so, the fixed point isn't on the boundary (from (3) scaled by the magnitude of x) so the result still follows easily.

In the same proposition, I wasn't able to follow the reasoning in your third argument which gives |x| < r (i.e. |x| is not r) without taking |y| < r/3 rather than |y| < r/2. This doesn't seem to affect anything adversely, though. I really like that argument since it makes the overall proof much more elementary.

I wasn't able to follow the reasoning in the last few lines–in particular replacing o(||x_n − x_*||) with o(t_n). I was able to show the result anyway, though I had to pass to a further subsequence and use the invertibility of the derivative to do so.

Minor typos/issues:

"point set" -> "point-set"

"A third argument (...) as follows" -> "A third argument (...) is as follows"

"image of any open neighborhood" -> "the image of any open neighborhood"

In the proof of Lemma 6, strictly speaking the B(y, r) should be closed in each case

In Lemma 7, dividing delta by 2 doesn't seem to serve a purpose

"then f(U_1) is non-empty contained in" -> "then f(U_1) is non-empty and contained in"

"conpactness" -> "compactness"

"This together with Corollary 11 with" -> "This together with Corollary 11"

In defining U' (and in the restatement right after), there's a missing vertical bar

[Corrected, thanks – T.]



Anonymous · 19 September 2011

What about the following proof? Is it related to Cernavskii's or Vaisala's?

As above, let's prove that f is a local homeomorphism at a point $x_0 \in \Omega$. We may assume $f(x_0) = x_0 = 0$ and $|f(x) - x| < \frac{1}{2}|x|$ for $x \in B(0,1)$.

Due to Proposition 3 above, for each $y \in \overline{B(0, \frac{1}{3})}$, there is a point in $x \in \overline{B(0, \frac{2}{3})}$ such that $f(x) = y$. Moreover, $f^{-1}(y)$ is compact. Take, in $f^{-1}(y)$, the point with maximum first coordinate. If there is more than one such point, take the that with maximum second coordinate, and so forth. Let the chosen point equal $g(y)$.

The function g is continuous, since

i) Because f is open, for every point near y there is a pre-image near $f(y)$.

ii) If there is a sequence $y_n \to y$ such that, say, $g(y_n)$ has first coordinate bigger than the first coordinate of $g(y)$ with a non-vanishing difference, we can take a subsequence y_{n_k} which converges to a point y' whose first coordinate is bigger than that of y, a contradiction.

g is obviously injective, and it is defined in a compact set. Thus, g is a homeomorphism.

This function is also a left inverse of f (that is, $g \circ f = id$ in the image of g). The claim then follows from the fact that injective continuous functions from n-dimensional sets to n-dimensional sets are open.



Terence Tao · 19 September 2011

I don't think the proof of the continuity of g is complete, even in the one-dimensional case. For instance, what prevents $g(y_n)$ from being significantly below $g(y)$ in one dimension?
Note that there are plenty of continuous proper maps without continuous left-inverses, e.g. $f(x) := x^3 - x$ in one dimension. (Note in this case that the function g as defined above is discontinuous.)



Anonymous · 19 September 2011

Errata: above, the line "which converges to a point y' whose first coordinate is bigger than that of y, a contradiction" is wrong. Instead of it, define $x = g(y)$ and read "such that $g(y_{n_k})$ converges to a point x' whose first coordinate is bigger than that of x, a contradiction".
Dear Tao,
indeed that "proof" is wrong. I still think the case $n = 1$ is ok:
Take any neighborhood N of x. If m is sufficiently large, $y_m \in f(N)$, so we have a candidate z_m for $g(y_m)$ in N. We may suppose $z_m \to x$ (e.g., by choosing z_m as close to x as possible).
In one dimension, by definition, x_m is the biggest element of $f^{-1}(y_m)$, so that its lim inf is at least $\lim_{m\to\infty} z_m = x$. Then, as above, if y_m does not go to x, we achieve a contradiction.
However, in more dimensions, I have overlooked something: the coordinates of the y_m's don't behave in the same way.
For the sake of clarity, given any $a, b \in \mathbb{R}^n$, let's say $a > b$ if and only if $a = b$ or one of the following conditions holds.
$a_1 > b_1$.
$a_1 = b_1$ and $a_2 > b_2$.
$a_1 = b_1$, $a_2 = b_2$ and $a_3 > b_3$.
Etc.
The failure of the proof is related to the instability of the relation above: if p and q are sequences such that $p_m > q_m$, $\lim_{m\to\infty} p_m = p$ and $\lim_{m\to\infty} q_m = q$, it is not necessarily true that $p \geq q$.
I will try to fix the argument, even though I think success is not probable.
Question: if g were continous, doesn't the conclusion follow? For it is known that injective continuous maps (in this case, g) are local homeomorphisms. So g would a homeomorphism between two neighborhoods of 0, and f should be its inverse.

Twelth Linkfest · 20 September 2011

[...] Tao: The inverse function theorem for everywhere differentiable maps, The Brunn-Minkowski inequality for nilpotent [...]



Anonymous · 21 September 2011

Dear Prof. Tao,
in the proof of lemma 7 you ask the reader to "note that $K_r \setminus K_r^1$ is closed". Can you give me a hint on how to see this? Components are closed in general (which gives the compactness of K_r^1), but I do not see, why K_r^1 is also open (relative to K_r).
Thanks in advance.



Terence Tao · 22 September 2011

Oops, the argument is not quite correct as stated; I've rewritten it. The key point is that in a compact set, any two points that don't lie in the same connected component can be separated from each other by clopen sets (but the connected components themselves need not be clopen).

Implicit function theorem | Mathitself · 8 October 2011
[...] deep result is that it's enough that is everywhere differentiable (see Tao's post), which remains one an other deep result called invariance of [...]

 **francescodifusco** · 9 March 2012
Reblogged this on FRANCESCO DI FUSCO.

 **Alan Macdonald** · 19 March 2012
The paper "A Strong Inverse Function Theorem" by William J. Knight
(The American Mathematical Monthly, Vol. 95, No. 7, pp. 648-651) gives an improvement of the standard result in a different direction.

 **Olaf Zurth** · 9 September 2012
A Question: Is there in the first line of Lemma 6 a typo?
$0, \frac{2}{10} \leq r \leq \frac{4}{10}$ should read as $\frac{2}{10} \leq r \leq \frac{4}{10}$ or did I misunderstood something?
[Corrected, thanks – T.]

 **blindman** · 8 November 2012
Dear Professor Tao,
the local inverses $f^{-1} : V \to U$ are also differentiable at x_0
should be replaced by
the local inverses $f^{-1} : V \to U$ are also differentiable at $f(x_0)$
[Corrected, thanks – T.]

 **blindman** · 8 November 2012
Dear Professor Terence Tao,
In the book "Mathematical Analysis on Manifold" of Michael Spivak, the author gave a counterexample of Theorem 1 (Exercise 2.39, page 52) if the assumption on the continuity of the derivate f' is violated.
They consider the function $f :: \mathbb{R} \to \mathbb{R}$ given by
+ $f(x) = \frac{x}{2} + x^2 \sin \frac{1}{x}$ if $x \neq 0$;
+ $f(x) = 0$ if $x = 0$
I would to ask your comments about this situation.
Thank your for your helping.

 **Terence Tao** · 8 November 2012
This function has non-zero derivative at 0, but has vanishing derivative at many other points (as can be seen for instance from a plot of the function
), and so does not contradict Theorem 1.

 **blindman** · 8 November 2012
Dear Sir. Thank you for your comments and helping.

 **pera** · 13 December 2012
And what would be counterexample?

 **Alexander** · 28 March 2014
Why not just to use the Mean value theorem on a segment $[x_1,x_2]$ to obtain injectivity immidiately instead of all these lemmas?

 **Alexander** · 29 March 2014

Ok, I see: because there's no such a theorem... sorry!

The inverse function theorem | Negro's notes · 14 December 2015

[...] differentiable conditions on a mapping which ensure that it is a local diffeomorphism. (But see Terry Tao's blog for a differentiable, non-smooth inverse function theorem). The main point of such theorems is the [...]

 **Mladen Đalto** · 24 November 2016

Is there a pseudoinverse version of this theorem for $ f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ where n is not equal m ?
If not, what method of proof/research would you recommend ?
I require it for machine learning research so any pointers would be appreciated.
Thank you.

 **Anonymous** · 15 April 2017

In Spivak's Calculus on manifolds, there is an exercise that the function $f : \mathbb{R} \to \mathbb{R}$ defined by
$f(x) = \frac{x}{2} + x^2 \sin \frac{1}{x}, \quad x \neq 0$
and $f(0) = 0$ shows that continuity of the derivative cannot be eliminated from the hypothesis of the inverse function theorem. Why does this not contradict Theorem 2?

>  **Terence Tao** · 16 April 2017
>
> This map f has an infinite number of critical points near the origin, so the hypotheses of Theorem 2 are not satisfied. (Perhaps the formulation of the inverse function theorem in Spivak is slightly different, in that one only demands invertibility of the derivative at a single point, rather than on an entire domain.)

 **Anonymous** · 17 April 2017

I'm very much curious about motivation of considering the map $x \mapsto x - f(x) + y$, which is the key step to recover the proof of Theorem 1. Other than just memorizing it, I have never seen in any textbook an explanation for a possible motivation (algebra? geometry? or something else) for this map.

>  **Terence Tao** · 17 April 2017
>
> If $f(0) = 0$ and $Df(0)$ is the identity, then Taylor expansion suggests the approximation $f(x) \approx x$ for small x. If this approximation were exact – $f(x) = x$ – then one could easily solve the equation $f(x) = y$ by writing it as $x = y$. But the approximation is inexact; the best one can do with regards to transforming $f(x) = y$ to an equation that looks like $x = y$ is to add $x - f(x)$ to both sides to obtain $x = x - f(x) + y$. The error here $x - f(x)$ is not zero, but it is "small", and in particular (for small x and continuously differentiable f) the map $x \mapsto x - f(x) + y$ still behaves somewhat like a constant, in that it is a contraction.

 **Joe Higgins** · 26 March 2018

Can you give a hint about the statement 'one can recover the differentiability of the local inverses' please? Much appreciated.

>  **Terence Tao** · 27 March 2018
>
> Let's say $y_0 = f(x_0)$. If y_n approaches y_0, then by the homeomorphism property we have local inverses $x_n = f^{-1}(y_n)$ that approach x_0. On the other hand, we have the Newton approximation

$f(x_n) = f(x_0) + Df(x_0)(x_n - x_0) + o(|x_n - x_0|)$. One can put these facts together to obtain the usual formula for the derivative of f^{-1} at y_0.

> **Anonymous** · 19 December 2023
>
> I'm a little confused about this, following the hint I obtain $f^{-1}(y) = f^{-1}(y_0) + (Df(x_0))^{-1}(y-y_0) + o(|f^{-1}(y)-f^{-1}(y_0)|$, so I have to prove that $o(|f^{-1}(y)-f^{-1}(y_0)|$ is also $o(|y-y_0|)$, but I can't find a way without supposing continuity of Df at the point x_0. Thank you in advance.

Jaikrishnan Janardhanan · 11 July 2020

There seems to be a new and elementary proof of the inverse function theorem for just everywhere differentiable maps:
https://www.jstor.org/stable/10.14321/realanalexch.43.2.0429

Eduardo Ramos · 11 January 2023

Hello Terence Tao. Are you still interested in such question? I think I solved the infinite dimension case, where I proved the same is true in Bannach Spaces as long as F is "locally proper" in some sense, which includes for instance functions of the form I+K, K being compact. Additionally I think the proof is actually simpler, and generates interesting results in more general spaces. Could you read my proof?

> **Eduardo Ramos** · 11 January 2023
>
> Specifically, I proved that for Banach spaces X and Y, if $F : U \subset X \to Y$ is Fréchet differentiable and $DF(x) : X \to Y$ invertible for all $x \in U$, then F will be a local homeomorphism if and only if F is 'locally proper'.
> Here F being locally proper means that for each $x_0 \in U$ there exists a closed neighborhood $V \subset U$ of x_0 such that if (x_n) is a sequence in V with $F(x_n) \to c \in Y$, then (x_n) has a convergent subsequence.

Eduardo Ramos · 21 January 2023

Well, after checking each detail of my proof I found an error. Actually with the above conditions I can only prove that F is an open map. Under the above hypothesis I also found a similar argument for the infinite dimensional case that can follow all steps of your arguments up to Proposition 12, where it sadly fails. Soon I will publish the open mapping result, and will keep working on the local injectivity problem.

A A · 20 July 2025

> ... uniform upper bound $\|f(x)\| < \frac{1}{10}$ for all $x \in \partial B(0, \frac{1}{20})$; ...

Dear Professor Tao — very minor and feel free to ignore, but think here you probably meant $x \in B(0, \frac{1}{20})$ (i.e the whole closed ball, not just its boundary).
[Corrected, thanks – T.]